MULTIPLICATION INDUCES ADDITION OF TANGENT VECTORS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma varieties-lemma-tangent-space-product (canonical tag 0BEB). cited at statement lines 911-912. Direct sum decomposition of the product tangent space.
- Lemma varieties-lemma-map-tangent-spaces (canonical tag 0B2F). cited at statement lines 913-914. The map of tangent spaces induced by m.
- Equation varieties-equation-tangent-space-fibre (canonical tag 0BEA). cited at proof line 918. Description of tangent vectors in fibres.

Source reference. Original lemma 0BEB: sources/stacks-chow/tags.txt, source lines 14344–14344. Cited at statement lines 911-912: identifies the direct sum of tangent spaces with the tangent space of the product; load-bearing.

Source reference. Original lemma 0B2F: sources/stacks-chow/tags.txt, source lines 13928–13928. Cited at statement lines 913-914: the right arrow is induced from m via this lemma; load-bearing.

Source reference. Original equation 0BEA: sources/stacks-chow/tags.txt, source lines 14343–14343. Cited at proof line 918: the proof works with tangent vectors in fibres via this equation; load-bearing.

Lemma 0.2. *Let S be a scheme. Let G be a group scheme over S. Let $s \in S$. Then the composition*

$$T_{G/S,e(s)} \oplus T_{G/S,e(s)} = T_{G \times_S G/S,(e(s),e(s))} \to T_{G/S,e(s)}$$

is addition of tangent vectors. Here the $=$ comes from Varieties, Lemma 0BEB and the right arrow is induced from $m : G \times_S G \to G$ via Varieties, Lemma 0B2F.

Proof. We will use Varieties, Equation (0BEA) and work with tangent vectors in fibres. An element θ in the first factor $T_{G_s/s,e(s)}$ is the image of θ via the map $T_{G_s/s,e(s)} \to T_{G_s \times G_s/s,(e(s),e(s))}$ coming from $(1, e) : G_s \to G_s \times G_s$. Since $m \circ (1, e) = 1$ we see that θ maps to θ by functoriality. Since the map is linear we see that (θ_1, θ_2) maps to $\theta_1 + \theta_2$. □

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